

6.849: Geometric Folding Algorithms

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These are my lecture notes from 6.849, Geometric Folding Algorithms, at the Massachusetts Institute of Technology, taught this semester (Spring 2017) by Professor Erik Demaine¹.

I wrote these lecture notes in L^AT_EX in real time during lectures, so there may be errors and typos. I have shamelessly pillaged Tony Zhang's² formatting commands. Should you encounter an error in the notes, wish to suggest improvements, or alert me to a failure on my part to keep the web notes updated, please contact me at rmwu@mit.edu.

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Contents

1	February 8, 2017	1
1.1	Administrivia	1
1.2	Introduction	1
1.3	Linkages	2
1.4	Paper	2
1.5	Polyhedra	2
1.6	Hinged dissections	2

1 February 8, 2017

1.1 Administrivia

- Lectures in 54-100, Wednesdays 2:30-5.
- Join the class mailing list, sign up for Coauthor, fill out the signup form.
- Watch video lectures from 2010, and we will solve problems in class. There are notes and slides too. Then fill out the video completion forms, and finish them by Tuesday noon.
- Weekly problem sets, relatively easy, choice of problems to solve. Upload pdfs to Gradescope, and submit folded paper physically.
- Physically attend class!
- Paper and presentation due at end of class. Typical projects include building stuff, code, posing open problems, research, or paper review. It's even okay to write a few Wikipedia articles about geometric folding.

You can also go to your favorite piracy site to find the textbook—edemaine.

1.2 Introduction

In this class, we are interested in mathematics and algorithms behind (un)folding geometric objects. We can also think of folding as reconfigurations.

There are many applications.

- In robotics, there are self-folding robots and programmable matter.
- In graphics, morphing shapes (polygons) is also a folding problem.
- In mechanical engineering, people have tried to draw straight lines by turning knobs. We can figure out how to manufacture without collisions.
- In medical fields, we could fold up small robots and tools. There is also the geometry of protein folding.
- In aero/astro, how do we fold up devices to fit into a space shuttle?
- Origami!

There are three types of objects: linkages, paper, and polyhedra. **Linkages** are one-dimensional line segments with hinges to connect them. They cannot cross, compress, or disassemble. **Paper** is two-dimensional, which can be folded as long as you don't stretch or tear the paper. **Polyhedra** are about building surfaces from two-dimensional shapes. Often, we care about unfolding polyhedra.

There are two types of problems: foldability/analysis (structure to foldings), and design/synthesis (foldings to structure). The former includes having a robot arm, and asking what it can do. The latter includes wanting to build an origami dragon, and designing how to make the folds.

There are three types of results: universality, decision, and complexity. Universality means that for some definition of "everything," everything can be folded.

Sometimes, not everything is foldable, so we have decision algorithms to say what is foldable. At times, we can also show that something is computationally hard to fold and intractable.

1.3 Linkages

One idea we consider is rigidity. It is easy (mathematically) to determine the boundary between rigid in 2D, flexible in 3D and flexible. It is hard to determine the boundary between the second category and rigid (which is concerning considering that we build so many rigid structures).

We consider universality, which asks which linkages can fold into all their configurations. In two dimensions, all chains are foldable. There's a cool demonstration. In three dimensions, that's not true! Proteins fold in three dimensions, and it's computationally intractable to fold in 3D. In four dimensions and higher, things are actually all foldable.

1.4 Paper

We ask, which crease patterns fold flat? Unfortunately, this is NP-hard (known 20 years ago). The natural design question is, which polyhedra can be folded from a square? For example, can we fold a horse from a piece of paper? How about a zebra from a two-side paper? How about a three-dimensional model?

Everything is foldable from a square piece of paper! There is an extremely inefficient algorithm to do so. There's also Origamizer.

Where's [my lecture notes]? Oh I folded it up—edemaine

Maze folding is another example, and so is folding with one straight cut. Demaine proceeds to do magic tricks with paper. (Holy shit this is so cool.)

1.5 Polyhedra

A cube can be unfolded by cutting along the edges. It can also be folded in other ways, and there's an algorithm to see what can be folded from that cross. Here, every point should be single covered.

There is edge unfolding and general unfolding (can cut anywhere you want). We can unfold convex polyhedron or general polyhedron.

	edge	general
convex	open	yes
general	no	open

1.6 Hinged dissections

Hinging is to cut up the parts of a shape and pivot on the surfaces. Given any two polygons, you can hinge one into the other.

1.7 Problem solving

Mountains are locally to the right (red), valleys locally to the left (left).